

CS 59300 Application of Deep Learning

Homework 7: Detecting Real and AI-Generated Synthetic Images

Due on **October 22, 2025, 11:59pm**

The goal of this homework assignment is to help you become familiar with convolutional neural networks (CNNs) for image classification. For this homework, please include all code in the same Jupyter Notebook, named “hw7.ipynb”. Ensure that all outputs are retained in the Jupyter Notebooks.

This assignment uses the CIFAKE dataset, which contains real and AI-generated synthetic images, available on Kaggle:

<https://www.kaggle.com/datasets/birdy654/cifake-real-and-ai-generated-synthetic-images>

The dataset includes 100,000 training images (50,000 per class) and 20,000 testing images (10,000 per class).

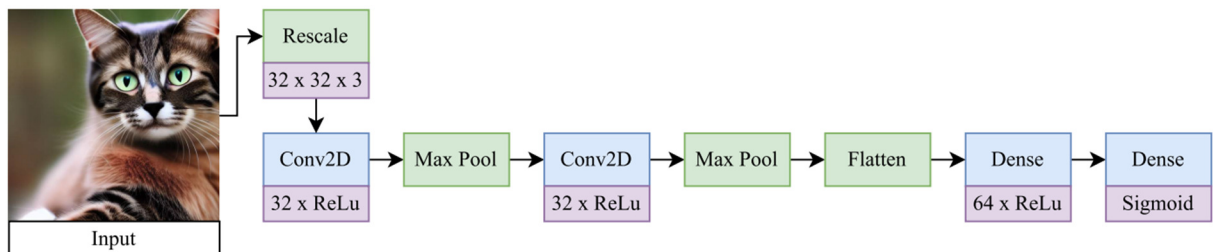
1. **Split the training dataset into a real training dataset and a validation dataset.** In the Jupyter Notebook “hw7.ipynb”, write Python code to accomplish the following:

- Download the dataset zipped file to Google Colab, such as using the “curl” command.
- Unzip the file. After unzipping, you will find two folders: “train” and “test”.
- Create a third folder named “validation”. Move 10,000 images from “train/REAL” to “validation/REAL” and 10,000 images from “train/FAKE” to “validation/FAKE”.

As a result, the train folder will contain 80,000 images, while the validation folder will have 20,000 images.

2. **Convolutional Neural Network (CNN):** Use the `image_dataset_from_directory()` function to load data from the “train”, “validation”, and “test” folders. Build the CNN model as described in the research paper “CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images” (IEEE Access) at

<https://ieeexplore.ieee.org/abstract/document/10409290>, i.e.,



which is the Figure 5 in the paper for the model architecture. Apply the “ModelCheckpoint” callback to save the best model based on validation loss during training, and train the model for 30 epochs. Ensure that your final model achieves at least 92% accuracy on the test dataset.

Please upload “hw7.ipynb” file to your GitHub homework repo under “hw7”.

Grading rubric:

- Part 1 – 20pts
- Part 2 – 80pts

Total: 100pts